

IB Portfolio, Explained

A plain-English guide to what you built, what the numbers mean, and how to answer when someone asks

This guide has one job: make sure that when an interviewer points at any cell in any of your models, you can say what it is, where it came from, and why you chose it. A portfolio you cannot defend is worse than no portfolio, because it turns a neutral first impression into a bad one in about ninety seconds.

It is written in ordinary language. Where a piece of jargon matters, it is explained the first time it appears and then used normally, because you will need to use it normally too.

How to use this

Read Part 1 once to fix the vocabulary. Read the Part 2 section for a project the night before you discuss it. Part 3 is the question bank — practise saying those answers out loud, not just reading them. If you can answer Part 3 without looking, you own the portfolio.

What is in here

Part	What it covers
1	The vocabulary — twelve terms that carry almost everything
2	Project by project — what each one is, in one line and then properly
3	The questions you will actually be asked, with your answers
4	The three rules that make this portfolio different, and why they matter

Figures throughout are the ones actually in your models, as of the 12 September 2026 market date used across the repository. If you rebuild a model and a number moves, update it here too.

Part 1 · The vocabulary

Almost all of investment banking valuation runs on about twelve ideas. Everything else is detail.

1. Equity value and enterprise value

Equity value is what the shareholders own: share price times number of shares. It is what you would pay to buy all the shares.

Enterprise value is what the whole business is worth regardless of who financed it: equity value plus debt, minus cash. The logic is that if you buy a company you inherit its debts, so you must pay for those too — but you also get its cash, so that comes off.

The one-line version

Buying a house for 1 million with a 300,000 mortgage attached and 50,000 in cash in the kitchen drawer: equity value 1 million, enterprise value 1,250,000. You pay the seller, you take on the mortgage, you keep the cash.

This matters constantly. Profit measures before interest (like EBITDA) belong to everybody — lenders and shareholders — so they pair with enterprise value. Profit measures after interest (like net income) belong only to shareholders, so they pair with equity value. Mixing them is the most common junior error.

2. EBITDA

Earnings before interest, tax, depreciation and amortisation. A rough measure of the cash a business generates from operations before financing and accounting choices get involved. Bankers like it because it lets you compare two companies with different debt loads and different depreciation policies.

It is not cash flow. It ignores the money spent on equipment and the money tied up in unpaid customer invoices. Anyone who tells you EBITDA is cash flow has not run a business.

3. Multiples

A multiple is a price expressed in years of profit. If a company is worth 10 billion and makes 1 billion of EBITDA, it trades at 10 times EBITDA. Saying “10 times” is quicker than saying the price, and it lets you compare a large company with a small one.

The two you will use most: **EV / EBITDA** and **P / E** (share price divided by earnings per share). The first values the whole business, the second values the shares only.

4. Discounted cash flow, or DCF

The idea: a business is worth the cash it will produce in future, but future cash is worth less than cash today, so you shrink each future year by a discount rate before adding them up.

Money today is worth more for two reasons: you could invest it, and the future is uncertain. The discount rate carries both.

5. WACC — the discount rate

Weighted average cost of capital: the blended return the company's investors require. If a business is funded half by shareholders wanting 12% and half by lenders charging 6%, its WACC is roughly 9%.

The shareholder half comes from CAPM: risk-free rate, plus beta times the equity risk premium.

- **Risk-free rate** — what the government pays to borrow for ten years. Your safest alternative.
- **Equity risk premium** — the extra return investors demand for owning shares instead of government bonds. Around 4.5% in developed markets; more where the country is riskier.
- **Beta** — how much this share moves when the whole market moves. Beta of 1 means it moves with the market; 0.9 means it moves a bit less; 1.5 means it swings harder.

6. Terminal value

A DCF forecasts five years in detail, then has to account for every year after that. The lump sum representing all of it is the terminal value. It is usually 60–80% of the answer, which is worth admitting rather than hiding: most of a DCF is an assumption about a distant year, not a forecast.

7. Free cash flow

The cash left after the business has paid its operating costs, its taxes, bought the equipment it needs, and funded the extra working capital that growth requires. It is what a DCF discounts, and what an LBO uses to repay debt.

8. Working capital

Money stuck in the business day to day — mostly customers who have not paid yet, less suppliers you have not paid yet. When a company grows, this number grows, and that growth consumes cash. That is why a profitable, fast-growing company can still run out of money.

9. Comparable companies and precedent transactions

Trading comparables ask: what is the stock market paying today for similar businesses? **Precedent transactions** ask: what have buyers actually paid to acquire similar businesses?

Precedents are almost always higher, because a buyer taking control pays a premium for it. The gap between the two is the control premium.

10. Accretion and dilution

In a takeover: does earnings per share go up or down for the buyer's existing shareholders? If up, the deal is accretive; if down, dilutive.

The mechanism is simple. The buyer gains the target's profits but usually issues new shares to pay for it. If profits rise by more than the share count, EPS rises. If not, it falls.

11. Tangible book value, and why banks are different

Book value is what the accounts say the shareholders own. **Tangible** book value strips out goodwill and other intangibles — things you cannot sell separately.

For banks this is the number that matters, because regulators require a bank to hold real capital against its loans, and goodwill does not count. A bank deal that destroys tangible book destroys the bank's capacity to lend, however good the earnings look.

12. IRR and MOIC

MOIC — multiple on invested capital. Put in 100, get back 200, that is 2.0 times.

IRR — internal rate of return: the same result expressed as an annual percentage. 2.0 times over five years is about 15% a year. Private equity funds typically target 20%+.

Why both

MOIC ignores time: doubling your money in two years and in ten years are both 2.0 times. IRR fixes that but flatters short deals. Bankers quote both because neither alone is honest.

Part 2 · Project by project

Twenty projects are built. Each section gives the one-line version, what it actually does, the numbers that came out, and the single thing that makes it worth showing to someone.

Project 1 · Apple — three-statement operating model

One line: a forecast of Apple's profit, balance sheet and cash, wired together so changing one assumption updates all three.

The three statements are the income statement (did we make money), the balance sheet (what do we own and owe) and the cash flow statement (where did the cash actually go). In a real model they are linked: profit flows into retained earnings on the balance sheet, and cash flow explains the change in the cash line.

The test of a three-statement model is that the balance sheet balances without you forcing it. If it does, your links are right. If you plug a number to make it balance, the model is decoration.

Yours has a scenario switch — one cell changes the whole model between base, upside and downside — and a revolver, which is an automatic overdraft that kicks in if the company would otherwise run out of cash.

Why it matters

This is the foundation model in banking. Everything else — DCF, LBO, merger — sits on top of one of these. Building one that ties is the basic proof that you can do the job.

Project 2 · Apple — discounted cash flow valuation

One line: what is Apple actually worth, based on the cash it will generate, rather than what the market says today.

Takes the forecast from Project 1, works out the free cash flow each year, and discounts it back at Apple's WACC. The WACC was built from scratch rather than assumed: ten peer companies, five years of weekly share price moves regressed against the market to get each one's beta, adjusted for how much debt each carries, then the median applied to Apple.

Item	Value	Plain English
Beta	1.12	Apple swings slightly harder than the market
Cost of equity	10.0%	What shareholders require to hold it
WACC	9.9%	Apple is almost all equity-funded, so the two are nearly the same

Why it matters

Terminal value is calculated two ways — by assuming perpetual growth, and by assuming a sale at an exit multiple — and each is checked against what the other implies. If assuming 3% growth forever implies selling at 25 times earnings, and you thought 12 times was right, one of your two assumptions is wrong. Most student DCFs never run that check.

Project 3 · TCS — trading comparables

One line: what the stock market pays for IT services companies today, and what that implies TCS is worth.

Eight peers across three tiers: Indian large caps, Indian mid caps, and global names like Accenture and Cognizant. For each, enterprise value is built up properly and four multiples calculated.

Two technical problems were solved here, and both are worth mentioning in an interview:

- **Currency.** Infosys reports its accounts in US dollars but its shares trade in rupees. Multiply those together carelessly and the multiple is wrong by a factor of 85. The model detects the reporting currency and normalises everything to dollars before any division happens.
- **Calendarisation.** TCS's year ends in March, Accenture's in August, Cognizant's in December. Comparing forward multiples across them compares different twelve-month periods. The model weights each peer's forecast years to line them all up with TCS's March year.

Why it matters

Anyone can pull multiples off a screen. Knowing that they are not comparable until you fix currency and fiscal year ends is the difference between a student and an analyst.

Project 4 · IT services — precedent transactions

One line: what buyers have actually paid for IT services businesses over ten years.

Fourteen deals from 2014 to 2024 — Capgemini buying IGATE, Atos buying Syntel, LTI merging with Mindtree, Cognizant buying Belcan and others — built from the original announcement press releases and filings rather than downloaded from a database.

Measure	25th percentile	Median	75th percentile
EV / revenue	1.8x	2.6x	3.4x
EV / EBITDA	13.6x	14.7x	16.9x
One-day premium	3.5%	4.6%	9.0%

Why it matters

The notes say plainly that part of the apparent control premium is not control at all — it is the cycle. Those deals were struck when the sector traded at 20–30 times earnings; it now trades at 10–15. Comparing deal multiples from one era with trading multiples from another overstates the premium. Spotting that is the kind of thing that gets remembered.

Project 5 · TCS — football field and valuation summary

One line: every method of valuing TCS on one chart, as ranges rather than single numbers.

A football field is the chart bankers put in front of clients: one horizontal bar per valuation method, each running from low to high. You read the overlaps. Where independent methods agree, you have something.

Method	Low	High	Weight
52-week trading range	1,977	3,350	0%
Analyst price targets (43)	1,775	3,900	10%
DCF	1,625	2,504	35%
Trading comps (three multiples)	1,242	2,689	45%
Precedent transactions	1,428	3,532	10%
Recommended range (weighted)	1,676	2,614	

All figures INR per share. The share price on the day was 2,201, which sits 2.6% above the weighted midpoint of 2,145.

Why it matters

Two things. First, the weights are inputs on the sheet, not hidden inside a formula — a client can argue with them, which is the conversation you want. Second, and better: the standard beta method from Project 2 was tried here and **rejected**. Regressing Indian shares against the S&P 500 produced betas of 0.06 to 0.33 with R-squared of 0.00 to 0.04 — meaning the relationship explains essentially nothing, because the two markets trade in different hours and different currencies. The regression was redone against the Indian market instead, giving 0.91 with R-squared 0.28. The rejected attempts are still on the sheet.

If you remember one thing for interviews

“R-squared” measures how much of one thing’s movement is explained by another, from 0 to 1. An R-squared of 0.01 means your beta is noise. Knowing to throw a result away is a stronger signal than producing one.

Project 6 · Emirates NBD / Mashreqbank — merger model

One line: should Emirates NBD buy Mashreqbank, and the answer is no.

A full accretion/dilution model with real purchase accounting. Bank deals are judged in a specific order, and the model follows it: tangible book dilution first, regulatory capital second, earnings third.

Test	Result	What it means
Price paid	2.16x tangible book	Emirates NBD’s own shares trade at 1.41x
EPS, year 3	(2.3%)	Still dilutive even at full synergies
Break-even synergies	44.6% of target costs	Against 25% assumed. Not achievable
Tangible book dilution	(16.7%)	Shareholders lose a sixth of their tangible book
Earnback	Never	Earnings never exceed standalone, so it is never recovered
Pro forma CET1	13.0%	Down 2 points; survives, but only just
Return on capital	8.6%	Against a 12% required return

Why it matters

An acquirer whose shares trade at 1.4 times tangible book, issuing those shares to buy assets at 2.2 times, hands value to the seller the moment the deal is announced. No amount of synergy phasing reverses it.

Rather than reporting “2.3% dilutive” and inviting an argument about whether synergies could be higher, the model solves for the level that would make it work: 44.6%. That closes the discussion. **A merger model that only ever says yes is not being used properly.**

CET1 is a bank’s core capital as a percentage of its risk-weighted loans. Regulators set a floor; here it is 11%. Goodwill is deducted in full, which is why an expensive bank acquisition eats capital.

Project 7 · Cognizant — leveraged buyout

One line: could a private equity firm take Cognizant private, and what is the most it could pay?

An LBO is buying a company mostly with borrowed money, using the company’s own cash flow to repay the debt, and selling it five years later. The return comes from three places: debt repaid, profit grown, and the multiple changing.

Item	Value
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Offer price (30% premium)	US\$81.36 per share
Entry multiple	9.58x EBITDA
Debt raised	5.25x EBITDA — US\$21.3bn
Sponsor equity cheque	US\$18.9bn
Exit at 9.00x in year 5	MOIC 2.00x, IRR 14.9%
Versus the 20% hurdle	Misses by 5.1 points
Maximum price at 20% IRR	US\$73.78 — an 18% premium

Why it matters

The exit multiple is set *below* the entry multiple deliberately. Assuming you sell for more than you paid is how an LBO model is made to produce whatever answer is wanted, and one of the sixteen checks fails the model if anyone raises it above entry.

The genuinely useful output is not the IRR. It is the maximum price — US\$73.78 — because that is the floor under the share price in any sale process. And it sits 18% above where Cognizant actually trades, which says the market is pricing it below what a leveraged buyer could justify.

Project 8 · Reliance Industries — debt capacity and capital structure

One line: how much more could Reliance borrow, and at what rating?

A rating agency does not look at the headline leverage a company reports. It adds back leases, deferred spectrum payments and other debt-like items, strips one-offs from EBITDA, and asks how much cash is left after capex to service debt. The model rebuilds Reliance's FY2026 balance sheet on that basis and asks how much capacity remains at each rating.

Measure	Result	What it means
Headline net debt / EBITDA	0.60x	What the company reports
Agency-adjusted net debt / EBITDA	1.25x	Leases and spectrum liabilities double it
Cash interest cover	5.0x	Comfortable, but not the 8x of a decade ago
Capacity at an A rating	Rs 44,272 crore	Funds-from-operations to debt is the binding test
Capacity at BBB	Rs 1,90,332 crore	Four times more room one notch down
WACC today / at optimum	12.88% / 12.84%	Barely any saving from more debt

Why it matters

The finding is that Reliance is already close to its optimal capital structure: the WACC curve is almost flat, so extra borrowing buys almost nothing. The useful output is the rating map — which test binds at each notch — because that is what a treasurer actually negotiates with. **Capacity is set by the weakest ratio, not the average.**

Project 9 · DEWA — rebuilding a real IPO

One line: was Dubai's largest IPO priced fairly, and what has happened since?

Takes the April 2022 flotation of Dubai's electricity and water utility and rebuilds the pricing from the prospectus: dividend yield, EBITDA multiple and DCF against the AED 2.25–2.48 range, the order book, the cornerstone investors and the upsizing from 6.5% to 18% of the company.

Item	Value
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Priced at	AED 2.48 — top of the range
Open-book demand	AED 315bn — 37x covered, reproduced
Cornerstones	AED 13.8bn, 62% of the deal
Price versus the midpoint of three methods	+7.6% — priced full
Day-one close	AED 2.87, +15.7% — AED 3.5bn 'left on the table'
Total return since, per year	6.2%, mostly dividends

Why it matters

An IPO that is 37 times covered and pops 16% looks underpriced. The model says it was priced above the midpoint of its own valuation methods and that the pop was scarcity: cornerstones took most of the deal and the free float was small. Four years on, the total return has been a utility's, not a growth stock's. **The order book measures allocation, not value.**

Project 10 · Reliance Industries — sum of the parts

One line: value each Reliance business on its own peers, add them up, and see what the market is paying for Jio.

Five segments — Jio, Retail, refining and petrochemicals, oil and gas, and others — each at the multiple its listed peers trade on, less net debt and minorities, less a holding-company discount.

Segment	Multiple	Peer anchor
Jio (digital)	12x EBITDA	Bharti Airtel 11.6x
Retail	25x	Between Walmart and DMart
Refining and chemicals	7.5x	Indian and US refiners
Result	Rs 1,065 per share	14.5% below the market
Market-implied Jio multiple	17.3x	49% premium to Bharti

Why it matters

The number to remember is not the SOTP but the implied multiple: to justify today's price the market must value Jio at 17 times EBITDA, half again what Bharti trades on, or Retail at 37 times. That reframes the conversation from 'is Reliance cheap' to 'is Jio worth a 49% premium to its only peer'. **A sum of the parts is most useful run backwards.**

Project 11 · Emirates NBD — valuing a bank

One line: banks are valued on book value and return on equity, not on EBITDA, and this is the method.

A dividend-discount model and an excess-return model, which are the same thing written two ways: a bank is worth its book value plus the present value of the returns it earns above its cost of equity. The cost of equity is built up from the US risk-free rate, a UAE country premium and a bank beta.

Item	Value
Cost of equity	10.0% (4.1% + 0.9% country + 1.0 × 5.0%)
Return on equity, fading 16.5% to 14%	Long-run growth 3%
Value per share	AED 41.71 against AED 30.98 — 35% upside
ROE the price implies	12.5%

Cost of equity the price implies	11.1%
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Why it matters

Either the market thinks ENBD's returns will fall to 12.5% — below anything it has earned since 2016 — or it discounts Gulf bank equity at a rate above what a country-risk build-up gives. The second is the honest reading, and it recurs in Project 19's bank screen. **For a bank, price-to-book is ROE against the cost of equity, and nothing else.**

Project 12 · Saudi Aramco — energy net asset value

One line: value the barrels under the Kingdom's fiscal terms and find the oil price the share price implies.

Oil companies are valued on the cash their reserves will produce. For Aramco the state's take dominates: a royalty that steps from 15% to 45% to 80% as Brent passes US\$70 and US\$100, and a 50% income tax. The model builds a per-barrel netback under those terms, runs it over the 52-year reserve life, and then across a price deck.

Item	Value
Share of the barrel Aramco keeps at US\$70	33 cents — US\$20.21 on US\$60.62
Share of a marginal dollar above US\$100	10 cents
NAV per share at US\$70, 8% discount rate	SAR 18.06 against SAR 25.56
Brent the price implies at 8%	US\$118
Discount rate the price implies at US\$70	5.7%
Brent at which the base dividend is covered	US\$68.3
Model rebuild of FY2025 free cash flow	Within 6% of reported

Why it matters

The shares are a claim on volume and cost, not on the oil price: above US\$100 the equity gets a dime of each dollar. The price is not an oil bet either; it implies a discount rate under 6%, which is how the market treats a quasi-sovereign yield stock. And the dividend, not the NAV, is the live question: it needs high-sixties Brent to be covered after a US\$52bn capex programme. **The calibration check — rebuilding last year's cash flow — is what earns the rest.**

Project 13 · Emaar Properties — developer net asset value

One line: a developer is worth its sales backlog, its rental assets and its land, not its earnings multiple.

Three blocks: the profit embedded in AED 155bn of sold-but-undelivered homes, the recurring income from malls and hotels at a yield, and the land bank at appraised values with a 50% haircut on the cash it will take to develop.

Item	Value
NAV per share	AED 20.23 against AED 11.80 — a 42% discount
Development block versus listed Emaar Development	Within 4.5%
Residual land value implied	AED 767 per square foot of buildable area

Why it matters

Developers trade at discounts to NAV everywhere, and the discount is the market's view of execution and cycle risk. The cross-check against the separately listed development arm is what makes the NAV credible: the model's value for that block lands within 5% of what the market pays for it directly. **When part of a sum-of-the-parts is listed, use it as the check.**

Project 14 · Dabur India — quality of earnings and working-capital peg

One line: what is the business really earning, and how much working capital does it need?

The report a buyer commissions before agreeing a price. It bridges from the headline EBITDA a seller quotes to the operating EBITDA a buyer should pay for, tests whether it turns into cash, and sets a normal level of working capital — the peg — so the buyer pays for the business and not for whatever sits on the balance sheet on closing day.

Item	Value
Headline EBITDA, FY2026	Rs 3,015 crore
Adjusted operating EBITDA	Rs 2,473 crore — 82% of headline
What the gap is	Interest and gains on a Rs 5,148 crore treasury book
What the gap is worth at 20x	Rs 10,800 crore — twice the cash pile itself
Payable days, FY2023 to FY2026	126 to 157
Working-capital peg / closing balance	Rs 366 crore / Rs 83 crore — buyer receives Rs 283 crore

Why it matters

Nearly a fifth of the headline EBITDA is income on cash. In a deal the cash transfers at face value; a buyer who also capitalises its income at 20 times has paid for the same asset twice. The working-capital question is smaller in money and larger in negotiation: whether a month of extra payable days is structural or unwinds after completion. **A QoE exists to find the number that is not in the accounts.**

Projects 15 and 16 · Salik — company profile, buyer screen and pitch book

One line: who could own Dubai's toll operator, what could they pay, and what would you pitch to the government?

Project 15 is the groundwork: four years of financials, trading against listed toll roads, twelve candidate buyers scored on five weighted criteria, and what a financial sponsor and a strategic operator could each pay. Project 16 is the twelve-slide pitch book built on it, with every number read from the Project 15 workbook so the two cannot disagree.

Item	Value
EV / EBITDA at the market	20.2x — a 30% premium to the peer median
Price return since the 2022 IPO	+169%
Top of the buyer screen	Vinci, Brookfield, KKR, Abertis
Sponsor ability to pay (6x debt, 18x exit, 13% IRR)	AED 4.66 — 13% below market
Strategic ability to pay (7% WACC, 45-year concession)	AED 6.05 — 13% above market
Pitch book recommendation	Place 10–15% with a sponsor-plus-sovereign anchor at a 5–12% premium

Why it matters

The screen's finding is who cannot buy: the government holds 75% and the regulator sets the tariff, so a change of control is not on offer and the best operators fall down the list on consent. A sponsor cannot reach the market price; only a strategic can justify a premium. The pitch follows the model rather than the other way round. **A deck that recommends what its own appendix cannot support is the commonest failure in banking.**

Project 17 · Kellanova / Mars — replicating a fairness opinion

One line: take a real proxy statement and see whether the bankers' valuation ranges can be rebuilt.

When Mars bought Kellanova for US\$83.50 a share, the proxy disclosed how Goldman Sachs and Lazard built their fairness opinions: projections, discount rates, multiple ranges, premia. The model takes only what was disclosed, back-solves the two undisclosed inputs from one range, and rebuilds the other eleven.

Analysis	Proxy range	Rebuilt
Goldman DCF	\$68.42–83.82	\$67.79–83.19
Lazard EBITDA comparables	\$57.20–73.80	\$57.01–74.10
Lazard P/E comparables	\$55.90–75.70	\$55.86–75.81
Both advisors' premia analyses	as disclosed	To the cent
Lazard DCF	\$69.60–79.10	\$61.07–78.76 — low end not reproducible
Offer against twelve ranges		Above 7, inside 5, below 0

Why it matters

Almost every range comes back within a dollar or two from a public document. The one that does not — the bottom of Lazard's DCF — needs a terminal cash flow close to NOPAT, which means Lazard normalised the terminal year and the proxy did not say so. The offer was a full price: top of one DCF range, above the other, above every trading multiple. **A fairness opinion is a set of defensible ranges around a price already negotiated; the replication shows how tightly they fit.**

Project 18 · Dubai World and Nakheel — restructuring

One line: the Gulf's largest restructuring rebuilt as liquidity, recoveries and debt-for-equity.

In November 2009 Dubai World asked for a standstill with US\$3.5bn due in three weeks against US\$0.5bn of cash. The model shows why a solvent group ran out of time, what each creditor class received in present value once maturities were pushed out five and eight years at 1% coupons, and what the government's US\$8.9bn conversion to equity was worth.

Item	Value
Asset cover over the plan, forced-sale values	0.93x with DP World, 0.47x without
Bank creditors' recovery, present value	71 cents — on a 0% nominal haircut
Value transferred from banks by the new terms	US\$2.7bn
Nakheel trade creditors (40% cash, 60% 10% sukuk)	98% — the best-treated class
Government recovery on its conversion, 2010 values	48%

Why it matters

Nobody took a nominal haircut except the government, and the banks took a real one: eight years at 1% when the market required 8% is 70 cents on the dollar, whatever the loan agreement says. Contractors who could stop the projects did better than lenders who could only sue. **In a restructuring, operational leverage beats legal seniority, and recovery is a present-value number.**

Project 19 · Deal-sourcing dashboard

One line: the Monday-morning tool — 52 companies across five exchanges, normalised, screened, and rebuilt on demand.

A Python script rather than a workbook, because the value is in the refresh. It pulls vendor data for Gulf, Indian and US names, converts every field in its own currency, recomputes each multiple from components, flags what the vendor got wrong, and runs four screens into an Excel dashboard and a chart panel.

Item	Value
Records flagged for a data problem	10 of 48 — one in five
Worst vendor error	Infosys, HCL at 956x and 1,218x (dollars over rupees); near 10x recomputed
Take-private screen	Cognizant, EPAM, Wipro
Banks below the justified price-to-book line	Dubai Islamic Bank, Emirates NBD, Mashreqbank, HDFC Bank

Why it matters

A comps refresh that trusts the vendor is not a refresh. The screen that found Cognizant for Project 7 is here as code, and a developer excluded from it because customer advances make EV/EBITDA meaningless for the sector. **Knowing when a metric does not apply is most of the job.**

Project 20 · HDFC Bank / HDFC Ltd — the capstone

One line: replicate India's largest merger at announcement, then check what happened.

Reuses the merger toolkit from Project 6 and the bank frame from Project 11 on the April 2022 deal, then does what a pitch never does: sets the announcement case against three years of reported results and the share price against the bank index.

Item	Value
Exchange ratio and premium	1.68x, 3.3%
HDFC Ltd holders' ownership / look-through share of profit, book	41% / 42%, 47%
Pro forma EPS before the drag / book accretion	−1.2% / +7.7%
Regulatory drag in year one	Rs 4,006 crore after tax — 29% of HDFC Ltd's profit
ROE: standalone FY2023 to FY2026	15.7% to 12.8%
Total shareholder return since the day before announcement	+3.8% against Nifty Bank +50.9%
Price / book, 2022 to today	3.48x to 2.05x; implied cost of equity 9.4% to 10.1%

Why it matters

The ratio was set on market value and favoured the bank. The accretion arithmetic was right: book value per share compounded at 12%. It was also the wrong question, because the same pro forma said ROE would fall three points, and a bank's price is book times a multiple set by ROE. The multiple halved; the price went nowhere; the cost of equity the market applies barely moved. **The de-rating was arithmetic, not sentiment — and it was visible on the back page of the pitch.**

Part 3 · The questions you will be asked

These are the questions an interviewer reaches for when they see this repository. Practise them out loud.

General

Q. Walk me through a DCF.

Forecast free cash flow for five years. Discount each year back at WACC. Calculate a terminal value for everything after year five, either by assuming perpetual growth or by applying an exit multiple, and discount that back too. Add them up to get enterprise value. Add net cash or subtract net debt to get equity value, then divide by shares for a per-share number.

Q. Why would you use EBITDA rather than net income?

EBITDA sits above interest and tax, so it is not affected by how a company is financed or where it is domiciled. That makes two companies with different debt loads comparable. The cost is that it ignores capital spending and working capital, so it is not cash flow.

Q. Which is higher, precedent transaction multiples or trading multiples, and why?

Precedents, normally, because an acquirer taking control pays a premium for synergies and for the right to run the business. In my Project 4 the gap is 2.6 times revenue against 1.8 times. But I would add that part of that gap is the cycle, not control — those deals were struck when the sector traded much higher, and comparing across eras overstates the premium.

Q. A company has negative free cash flow but is profitable. How?

Usually working capital. If it is growing fast, customers owe it more each month and inventory rises, and that swallows cash before it ever reaches the bank. Heavy capital spending does the same.

Project 2 — the Apple DCF

Q. Where did your WACC come from?

Bottom-up. I took ten peers, regressed five years of weekly returns against the S&P 500 to get each one's beta, unlevered each at its own net debt to equity, took the median, then relevered at Apple's capital structure — which is zero, because Apple is net cash. Then Blume-adjusted, which is two-thirds raw beta plus one third, the way banks do it. That gave 1.12, a cost of equity of 10.0% on a 4.95% ten-year Treasury and a 4.5% equity risk premium, and a WACC of 9.9%.

Q. Why is your WACC almost the same as your cost of equity?

Because equity is about 98% of Apple's capital. There is barely any debt to weight in.

Project 5 — the TCS football field

Q. Why is your TCS discount rate 13.3% when Apple's was 9.9%?

Different country and different currency. TCS's cash flows are in rupees, so I discount at a rupee cost of capital: a 7.02% Indian ten-year government bond rather than a 4.95% US Treasury, and an equity risk premium of 6.7% rather than 4.5%, because that adds India country risk. Discounting rupee cash flows at a dollar rate would bury an exchange-rate forecast inside the valuation.

Q. Why 5.5% terminal growth?

Roughly Indian long-run inflation plus a little real growth, and comfortably below India's nominal GDP growth — nothing can grow faster than its economy forever. I also cross-check it: 5.5% implies a 9.1 times exit multiple, and the 11 times exit multiple implies 6.8% growth. The two are close enough that neither contradicts the other.

Q. Your DCF says 1,953 and the stock is 2,201. Is TCS a sell?

I would not put it that strongly. My weighted range is 1,676 to 2,614 and the price sits inside it, about 2.6% above the midpoint. The honest reading is that the market is paying a modest premium to both my intrinsic value and the peer group. Whether that is wrong depends on one specific question — whether the sector's AI de-rating is permanent — and my model does not answer that.

Q. Seventy per cent of your DCF is terminal value. Isn't that a problem?

It is a limitation and I say so on the sheet. It is normal for an asset-light business with low capital intensity, but it does mean most of the answer is an assumption about 2031. That is exactly why I show the DCF as a sensitivity range on the football field rather than as a single point.

Project 6 — the bank merger**Q. Why do banks care about tangible book value more than EPS?**

Because regulatory capital is built from tangible equity. Goodwill is deducted in full from CET1, so paying a large premium for a bank converts real capital into an intangible that cannot support any lending. A deal can look accretive to earnings and still leave the bank less able to write loans.

Q. Your model says the deal fails. Why build it?

Because that is the answer. Emirates NBD trades at 1.4 times tangible book and would be paying 2.2 times. Rather than stop at 'dilutive', I solved for the synergies needed to break even — 44.6% of Mashreq's cost base, roughly double what is assumed and above anything disclosed in comparable bank mergers. In a real process that number is what you take to the client.

Q. Why did you set revenue synergies to zero?

Because they are the standard way this model gets made to work — a cross-sell assumption big enough to close the gap with no mechanism behind it. I left the line in at zero so the choice is visible. It would not rescue this deal anyway; the shortfall is about AED 700m after tax.

Q. Why is the consideration 70% stock?

Capital. At that mix pro forma CET1 lands at 13.0%, about two points above the minimum. A cash-heavy structure would not clear the floor, because cash paid away does not come back as capital.

Project 7 — the LBO**Q. What drives returns in an LBO?**

Three things: repaying debt with the company's own cash flow, growing EBITDA, and any change in the multiple between buying and selling. In mine the bridge is roughly US\$11.4bn from debt paydown and US\$11.2bn from EBITDA growth, less US\$2.4bn lost to multiple contraction.

Q. Why did you assume a lower exit multiple than entry?

Because assuming multiple expansion is how an LBO model is made to say yes. If the return only works when you sell higher than you bought, you have not analysed anything. I also put a check in the model that fails if anyone later raises the exit multiple above entry.

Q. Why does interest use the opening balance rather than the average?

Average balances make the model circular — interest drives cash flow, cash flow drives the sweep, the sweep drives interest. Resolving that needs iterative calculation enabled, and a workbook that only computes correctly with a setting switched on will break on someone else's machine. Using the opening balance slightly overstates interest, which is conservative.

Q. What is a cash sweep?

A term in the loan requiring spare cash to go toward repaying debt rather than to the owners. Mine applies 100% of free cash flow to the term loan until it is repaid. The notes are bullet, meaning they cannot be repaid early and are settled at maturity.

Q. Could a fund really write an US\$18.9 billion cheque?

Not alone. It would be a consortium. That changes who signs, not the arithmetic, and I say so in the model.

Projects 8 to 20

Q. Why is your Aramco NAV 30% below the share price? Are you saying it is overvalued?

No. At an 8% discount rate and US\$70 Brent the reserves are worth SAR 18 a share, but the same model says the price implies a 5.7% discount rate, and that is the credible reading: the state owns most of the float and the market treats the stock as a quasi-sovereign yield instrument. The model reports both. What I would defend is the netback: it rebuilds Aramco's reported free cash flow within 6%.

Q. In the Dabur QoE you remove interest income from EBITDA. Isn't that income real?

It is real, and the buyer gets the cash that earns it at face value on closing. What the buyer should not do is also pay a 20 times multiple for the income on cash it has just bought at one times. Removing it from EBITDA and adding the cash to the price is the standard treatment; leaving it in double-counts Rs 5,000 crore of cash.

Q. Your Salik pitch recommends a minority stake, not a sale. Isn't a sale the bigger fee?

The model says no sponsor can reach the market price and the government will not sell control of a tariff-setting asset. A pitch that recommends a sale of control would contradict its own appendix. The credible mandate is the anchor placement with an operator partnership behind it, and a banker who pitches the deal that can happen wins the one after it.

Q. You could not reproduce Lazard's DCF low end. Doesn't that undermine the replication?

It is the most useful result in it. Eleven of twelve ranges rebuild within a dollar or two. The one that does not needs a terminal cash flow close to NOPAT rather than the projected free cash flow, which tells you Lazard normalised the terminal year and the proxy did not disclose it. I would rather report that than force a fit.

Q. In Dubai World the banks got 100 cents. Why do you say 71?

Because five to eight years at a 1% coupon, when comparable paper yielded 8%, is worth 71 cents on the dollar today whatever the face value says. The banks accepted it because regulatory accounting let them hold the paper near par, and that is exactly why they would not take equity. Nominal recovery and economic recovery are different numbers, and a restructuring adviser prices the second.

Q. HDFC Bank's merger was accretive to book and EPS. Why has the stock done nothing?

Because a bank is priced on book times a multiple set by ROE, and the merger cut ROE by three points while growing the book. The multiple went from 3.5 to 2 times; the cost of equity the price implies barely moved. Everything the announcement promised happened; it just was not what the price depends on. The deal will be judged on whether deposits replace HDFC Ltd's bonds fast enough to rebuild the return.

Part 4 · The three rules

These are what separate this repository from the hundreds of others with the same project names on them. Be able to say why each one matters.

1. Every number is a live formula, and a checks sheet proves it

No answer is typed in. Every model has a Checks sheet that has to read MODEL OK, testing things like: does the balance sheet balance, does enterprise value equal the sum of its parts, does the money multiple agree with the IRR, is every forecast cash flow positive.

Some checks are not arithmetic at all — they are discipline. Project 5 fails if forecast revenue growth ever exceeds anything TCS achieved in the last three years. Project 7 fails if the exit multiple is set above entry. Those exist because quietly lifting an assumption until the answer looks right is the most common way a model goes wrong, and a check is cheaper than a reviewer.

2. Every source is named, and the weak ones are flagged

Where a number is approximate, it says so. Project 4 marks which target revenues came from rounded press releases. Project 6 flags that the credit mark is an assumption on an estimated loan book and the largest judgement in the model. Project 5 states that no broker-target bar could be verified from a free source.

Admitting the weak spots before someone finds them is what makes the strong parts believable.

3. Every project says what it cannot do

Each notes page has a section on what it would take to do the analysis properly, and what the analysis does not support. Project 4 says its TCS range is illustrative because no target in the set is remotely TCS's size. Project 5 says 70% of the DCF is terminal value. Project 7 says the leverage sensitivity flatters high-leverage cases and explains why.

The point of all three

An interviewer is not checking whether your model is right. They are checking whether you know where it is wrong. Every senior banker has seen a confident model blow up. What they are hiring for is judgement about limits, and this portfolio is built to demonstrate exactly that.

Before any interview

- Open the model you are about to discuss and click through the Checks sheet. Know what each check tests.
- Know your three headline numbers cold: WACC, the valuation conclusion, and the one assumption the answer is most sensitive to.
- Be ready to say what you would do differently with more time. It is the question most candidates fumble and it is written on every notes page.
- Never claim a number you cannot source. 'I would have to check that' costs you nothing; a wrong figure stated confidently costs you the interview.